

Enterprise Table Pattern

Version: 1.0
Component: DataTable
Last Updated: 2025-01-XX

Overview

The Enterprise Table Pattern defines standards for data tables across all TPI portals. Tables are the primary data display mechanism for lists, reports, and management interfaces.

Column Density Rules

Density Levels

Compact (40px row height)

- Use when: Displaying many rows (>100), space-constrained views
- Cell padding: 8px vertical, 12px horizontal
- Font size: 12px (text-xs)
- Icon size: 16px

Standard (48px row height)

- Use when: Default for most tables
- Cell padding: 12px vertical, 16px horizontal
- Font size: 14px (text-sm)
- Icon size: 20px

Spacious (56px row height)

- Use when: Important data, accessibility priority, touch interfaces
- Cell padding: 16px vertical, 20px horizontal
- Font size: 14px (text-sm)
- Icon size: 24px

Column Width Rules

- **Auto-width:** Text columns (name, description)
 - **Fixed-width:** Numeric columns (120px), status (100px), actions (80px)
 - **Flexible:** Primary content column (min-width: 200px, flex: 1)
 - **Min-width:** 80px per column
 - **Max-width:** 400px per column (truncate with ellipsis)
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Default Sort Rules

Sort Priority

1. **Most Recent First** (default for time-based entities)
 - Column: **created_at** or **updated_at**
 - Direction: Descending
 - Use for: Incidents, tasks, logs, activities
2. **Alphabetical** (default for reference entities)
 - Column: **name** or **title**
 - Direction: Ascending
 - Use for: Users, services, configurations
3. **Status Priority** (default for workflow entities)
 - Column: **status** (custom sort order)
 - Direction: Custom (Active → Pending → Resolved)
 - Use for: Work items, approvals, requests
4. **Numeric** (default for metric entities)
 - Column: Primary metric
 - Direction: Descending (highest first)
 - Use for: Performance data, rankings

Sort Indicators

- **Unsorted**: No indicator
 - **Ascending**: ↑ icon, primary color
 - **Descending**: ↓ icon, primary color
 - **Multi-sort**: Show all sorted columns with indicators
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Bulk Actions + Selection UX

Selection Behavior

Single Selection

- Click row: Selects row (if selection enabled)
- Checkbox: Toggle selection
- Keyboard: Space bar toggles selection

Multi-Selection

- Checkbox column: First column (fixed 48px width)
- "Select All" checkbox: In header
- Selected count: "3 selected" badge in toolbar
- Selection persists: Across pagination (if same filter)

Bulk Actions Toolbar

Toolbar Appearance

- Shows when: 1+ items selected
- Position: Above table, replaces filter bar temporarily
- Sticky: Sticks to top on scroll

Toolbar Content

[3 selected] [Bulk Action 1] [Bulk Action 2] [Clear Selection]

Bulk Action Types

- Delete: Destructive, requires confirmation
- Export: Export selected items
- Update Status: Change status of selected items
- Assign: Assign to user/team
- Archive: Move to archive

Bulk Action Flow

1. User selects items
2. Toolbar appears
3. User clicks bulk action
4. Confirmation dialog (if destructive)
5. Action applied to all selected
6. Success toast: "3 items updated"
7. Selection cleared

Saved Views

View Management

Save Current View

- Button: "Save View" in filter bar
- Dialog: Name input + "Save as default" checkbox
- Saved: Filters, sort, column visibility, page size

Load Saved View

- Dropdown: "Saved Views" in filter bar
- List: All saved views + "Default View"
- Actions: Load, Rename, Delete, Set as Default

Default View

- One default view per user per table
- Auto-applied: On page load

- Override: User can change and save new default

View Sharing (optional)

- Share view: Generate shareable link
- Permissions: Role-based view access
- Use case: Team-standard filters

Row Click => DetailDrawer

Click Behavior

Row Click Action

- Primary action: Opens DetailDrawer
- Drawer content: Entity detail (Overview tab)
- Drawer width: 600px (standard), 800px (wide)

Row Hover State

- Background: Subtle highlight (bg-muted/50)
- Cursor: Pointer
- No other visual changes

Disabled Row Click

- Use when: Row is not actionable
- Visual: Reduced opacity, no hover
- Example: Archived items, read-only data

Drawer Integration

Drawer Content Structure

```
DetailDrawer
├─ Header (title + close)
├─ Tabs (Overview / Activity / Audit)
└─ Footer (actions, if editable)
```

Drawer Behavior

- Opens: Slide-in from right (300ms)
- Closes: Slide-out to right (300ms)
- Backdrop: 50% opacity, clickable to close
- Focus: Trapped in drawer when open

Drawer Actions

- Edit: Inline edit form (if permitted)

- Delete: Opens ConfirmDialog
 - Export: Export entity data
 - Share: Copy link to entity
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Accessibility

Keyboard Navigation

Table Navigation

- Tab: Move between interactive elements
- Arrow keys: Navigate cells (if enabled)
- Enter/Space: Activate row (open drawer)
- Escape: Close drawer/modal

Sorting

- Tab to header
- Enter: Toggle sort
- Shift+Enter: Add secondary sort

Selection

- Space: Toggle row selection
- Shift+Space: Select range
- Ctrl/Cmd+A: Select all (if enabled)

Screen Reader Support

ARIA Labels

- Table: `aria-label="[Entity] table"`
- Sortable header: `aria-sort="ascending|descending|none"`
- Selected row: `aria-selected="true"`
- Row count: "Showing 1-25 of 247 [entities]"

Announcements

- Sort change: "Sorted by [column] [direction]"
- Selection: "[N] items selected"
- Filter: "Filtered to [N] results"

Focus Management

Focus Indicators

- Visible: 2px solid primary color outline
- High contrast: Meets WCAG AA
- Always visible: Never remove focus styles

Focus Order

1. Filter bar
 2. Table header (sort controls)
 3. Table rows (left to right, top to bottom)
 4. Pagination
 5. Actions
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Export Behavior

Export Formats

CSV Export (default)

- Includes: All visible columns
- Scope: Current filtered/sorted view
- Filename: `[entity]_[date]_[time].csv`
- Headers: Column labels (not IDs)

PDF Export (optional)

- Format: Formatted table with branding
- Includes: Filters applied, export date
- Filename: `[entity]_[date]_[time].pdf`
- Pagination: Multiple pages if needed

XLSX Export (optional)

- Format: Excel-compatible
- Includes: Multiple sheets (if applicable)
- Filename: `[entity]_[date]_[time].xlsx`

Export Flow

1. User clicks "Export" button
2. Dropdown: Format selection (CSV, PDF, XLSX)
3. User selects format
4. Loading: Button shows spinner
5. Download: File downloads automatically
6. Success: Toast notification "Exported 247 items"

Export Scope

What's Exported

- Current filtered data
- Current sorted order
- Visible columns only
- Selected items (if bulk export)

What's Not Exported

- Hidden columns
 - Pagination (all matching rows, not just current page)
 - Computed/derived columns (unless specified)
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Performance Guidelines

Virtualization

When to Virtualize

- Tables with >100 rows
- Complex cell rendering
- Real-time updates

Virtualization Rules

- Render: Visible rows + 5 buffer rows
- Scroll: Smooth, 60fps target
- Height: Fixed row height required

Data Loading

Progressive Loading

1. Load first page (25 rows)
2. Render table immediately
3. Load remaining data in background
4. Update pagination when ready

Infinite Scroll (optional)

- Alternative to pagination
 - Load more on scroll to bottom
 - Show loading indicator
 - Use for: Activity feeds, logs
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Table States

Loading State

- Skeleton rows: Match table structure
- Count: Show 5-10 skeleton rows
- Animation: Shimmer effect

Empty State

- Component: EmptyState
- Message: Contextual (no data vs. filtered empty)
- Action: Primary CTA if applicable

Error State

- Component: ErrorState
- Message: User-friendly error
- Action: Retry button

Partial Data State

- Show: Available data
 - Indicator: "Some data may be unavailable"
 - Use when: Partial API failure
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Column Configuration

Column Visibility

Show/Hide Columns

- Button: "Columns" in table toolbar
- Menu: Checkboxes for each column
- Persist: Saved to user preferences
- Required: At least 1 column visible

Column Reordering

Drag to Reorder (optional)

- Drag handle: Left edge of header
- Visual: Column highlights on drag
- Persist: Saved to user preferences
- Reset: "Reset to default" option

Column Resizing

Resize Handles (optional)

- Handle: Right edge of header
 - Visual: Cursor changes to resize
 - Min-width: 80px
 - Max-width: 400px
 - Persist: Saved to user preferences
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Best Practices

Do

- ☒ Use standard density for most tables
- ☒ Provide default sort that makes sense
- ☒ Enable row click for detail view

- ☒ Show selection count in toolbar
- ☒ Export includes current filters
- ☒ Virtualize large tables
- ☒ Provide keyboard navigation
- ☒ Announce state changes to screen readers

Don't

- ☒ Mix densities in same table
- ☒ Sort by non-meaningful columns by default
- ☒ Require hover to see important data
- ☒ Hide bulk actions behind menus
- ☒ Export without user confirmation (for large datasets)
- ☒ Render 1000+ rows without virtualization
- ☒ Remove focus indicators
- ☒ Use color alone to indicate status

Implementation Example

```
<DataTable
  data={incidents}
  columns={[
    {
      id: "number",
      header: "Number",
      accessor: (row) => row.number,
      sortable: true,
      width: 120
    },
    {
      id: "title",
      header: "Title",
      accessor: (row) => row.title,
      sortable: true,
      render: (value, row) => (
        <div>
          <div className="font-medium">{value}</div>
          <div className="text-xs text-muted-foreground">{row.description}</div>
        </div>
      )
    },
    {
      id: "status",
      header: "Status",
      accessor: (row) => row.status,
      sortable: true,
      render: (value) => <StatusBadge status={value} label={value} />,
      width: 100
    },
  ]
}
```

```
        id: "created_at",
        header: "Created",
        accessor: (row) => row.created_at,
        sortable: true,
        render: (value) => formatDate(value),
        width: 150
      }
    ]}
  pagination={{
    page: currentPage,
    pageSize: 25,
    total: totalIncidents,
    onPageChange: setPage,
    onPageSizeChange: setPageSize
  }}
  sorting={{
    column: "created_at",
    direction: "desc",
    onSort: handleSort
  }}
  selection={{
    selected: selectedIds,
    onSelectionChange: setSelectedIds
  }}
  onClick={row => openDrawer(row.id)}
  exportable
  onExport={handleExport}
  density="standard"
  emptyState={
    <EmptyState
      title="No incidents found"
      description="Create your first incident to get started."
      action={{ label: "New Incident", onClick: handleCreate }}
    />
  }
/>
```

Related:

- [Drawer Pattern](#)
- [Component Inventory](#)